



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

March 04, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 8927
(THIRTEENTH REVISION)

EXPIRATION DATE: 2026-02-28

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Meggitt Safety Systems, Inc.
Simi Valley, CA
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of certain Division 2.2 hazardous materials in non-DOT specification, small, high pressure spheres of welded construction, for military weapons use only. This special permit provides no relief from any Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. Party status will not be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a in that non-DOT specification packaging is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Meggitt Safety Systems, Inc., dated February 19, 2022 and submitted in accordance with § 107.109.

Tracking Number: 2022024789

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Helium, compressed	2.2	UN1046	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Prescribed packaging is a nonrefillable non-DOT specification girth-welded steel sphere in full compliance with DOT Specification 3HT except as follows:

178.44(a) - Type, size and service pressure.

The sphere may be girth-welded and have one end fitting welded as shown in HTL drawing 32197872 Rev. O dated September 3, 1982 or drawing 32197943 Rev. D dated January 3, 1986. Maximum water capacity is 46 cubic inches with a maximum service pressure of 8,000 psi.

178.44(b) - Authorized steel.

Type ARMCO Nitronic 40 (21-6-9) austenitic stainless steel with the following composition is authorized with proper welding procedure.

<u>CHEMICAL ANALYSIS</u>	<u>Percent</u>
Carbon	0.040 max.
Manganese	8.00/10.00
Phosphorus	0.060 max.
Sulfur	0.030 max.
Silicon	1.00 max.
Chromium	19.00/21.50
Nickel	5.50/7.50
Nitrogen	0.15/0.40

TYPICAL MECHANICAL PROPERTIES

<u>Condition</u>	<u>Ultimate Tensile (PSI)</u>	<u>Yield Strength (PSI)</u>	<u>Elongation Percent In 2 inches</u>
Annealed	112,000	68,000	44
20% Cold Reduction	150,000	120,000	18

178.44(d) - Manufacture.

(Added) Each sphere must be subjected to a process treatment after welding and before stress relieving by hydrostatically pressurizing to at least 100 percent but not more than 110 percent of the test pressure, and maintained at this pressure for 3 minutes. The process treatment must be witnessed by the independent inspector. Total and permanent expansion for this process treatment need not be recorded.

178.44(e) - Welding or brazing.

Welding as prescribed in § 178.44(a) of this special permit, is authorized. All pressure welds must be 100 percent radiographed after hydrostatic test.

178.44(f) - Wall thickness.

(1) Applies except that the minimum wall thickness must be such that the wall stress at the minimum specified test pressure must not exceed 116,500 psi.

(2) and (3) Not applicable.

(4) (Added) Calculations for spheres must be made by the formula:

$$S = PD/4tE; \text{ where}$$

E = 0.85 which must be applied to the girth weld area and the heat affected zones which zones must extend a distance of at least 6 times wall thickness from the center of the weld.

E = 1.0 for all other areas

178.44(g) - Heat treatment.

The half spheres may be stress relieved or annealed for forming. Welded spheres must be stress relieved at a temperature of 900 °F $\pm$ 25 °F for one hour. Stress relieving is to be done after process treatment, and before hydrostatic test.

178.44(j) - Cycling test.

Applies except the test on each lot is not required. Results of design qualification testing per § 178.44(p)(4) of this special permit must be submitted to the Office of Hazardous Materials Safety (OHMS) prior to initial shipment.

178.44(l) - Flattening test.

Test one sphere taken at random from each lot by flattening between parallel steel plates at the weld with the welded seam at right angle to the plate.

178.44(m) - Physical tests - Not required.

178.44(n) - Magnetic particle inspection.

Not required. Instead, each sphere must be inspected using apparatus and procedures for liquid penetrant examination in accordance with ASTM-E-165-65. Inspection must be performed externally on the finished container after the hydrostatic test. Evidence of discontinuities, which in the opinion of the independent inspector may appreciably weaken or decrease the durability of the sphere, must be cause for rejection.

178.44(p) - Acceptable results of tests.

- (1) Flattening required without cracking to half the diameter of the sphere.
- (2) Physical tests: Not required.
- (3) Burst pressure must be at least 4/3 times the test pressure. Actual burst pressure must be recorded.
- (4) Cycling: The design must be qualified by cycling 3 spheres from zero to service pressure to at least 35,000 pressurizations without evidence of distortion or failure.

178.44(q) - Rejected spheres.

- (1) Repair of welded seams by welding prior process treatment authorized; subsequent thereto containers must be heat-treated and pass all prescribed tests.

- (2) For each cylinder subjected to reheat treatment during original manufacture, sidewall measurements must be made to verify that the minimum sidewall thickness meets specification requirements after the final heat treatment.

178.44(r) - Marking.

Spheres must be marked by electro-etching. Instead of DOT-3HT, cylinders must be marked "DOT-SP 8927" followed by the service pressure. Stamping of elastic expansion is not required. Nameplates are not authorized.

b. OPERATIONAL CONTROLS -

- (1) Approval of the pressure relief device required by § 173.301(f) must be submitted to the OHMS prior to initial shipment.
- (2) The spheres are restricted to use in military weapons systems.
- (3) The spheres must be shipped in strong outside packagings in conformance with § 173.301(a)(9).
- (4) The spheres may not be refilled and reshipped.

8. SPECIAL PROVISIONS:

- a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modifications or changes are made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.
- b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo aircraft only, and passenger-carrying aircraft.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) — “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notices of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: ae